EFU-W Complete Module Package (Zenodo-Ready, Polished English Version)

EFU-W Complete Module Package

050.1 | 050.1.5 | 050.1.6 | 050.2 | v1.0 FINAL

EFU SYSTEM v5.1 – ZENODO UPLOAD PACKAGE
Wellbeing Flux Efficiency Index + Epigenetic Memory Index
Complete Module Set – Upload Ready

Author: István Simor
Date: 2026-03-22
Status: v1.0 FINAL
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Repository: github.com/x8cqjnbvnm-hub/EFU-System-v5.1-Diamond

Related DOIs:
10.5281/zenodo.18888082
10.5281/zenodo.18766850

FILE 1 – Zenodo Record Metadata (Submission Form)

FIELD	CONTENT
Resource type	Publication → Technical note
Title	**EFU-W: Wellbeing Flux Efficiency Index and Epigenetic Memory Index — Dimensionless Modulators for Biophysical Civilizational Accounting (EFU System v5.1, Modules 050.1–050.2)**
Authors	Simor, István [ORCID: to be inserted]
Publication date	2026-03-22
Description	[→ FILE 2: Abstract text]

FIELD	CONTENT
Keywords	embodied flux unit, EFU system, wellbeing quantification, biophysical economics, η(W), epigenetic memory, telomere length, NR3C1 methylation, FKBP5, HRV normalization, CEWS, civilizational stability, asymmetry theorem, antiparasitic resistance, regenerative protocols
License	Creative Commons Attribution 4.0 International (CC BY 4.0)
Version	1.0
Language	English

Related identifiers

- **Is part of:**
  github.com/x8cqjnbvnm-hub/EFU-System-v5.1-Diamond
- **Is related to:**
  DOI 10.5281/zenodo.18888082 (EFU System v5.1 main repository)
- **Is related to:**
  DOI 10.5281/zenodo.18766850 (MROI preprint — companion metric)

Funding

Independent research — no institutional funding

Notes

Module 050.3 (integrated η(W)_effective combining 050.1 × 050.2) is currently in preparation and will be released as v1.1 following pilot calibration.

FILE 2 – Abstract (Zenodo Description Field)

The EFU (Embodied Flux Unit) System quantifies civilizational metabolism through biophysical flux accounting. This technical note introduces two interdependent modules that complete the human wellbeing dimension of the framework.

MODULE 050.1 — WELLBEING FLUX EFFICIENCY INDEX η(W)

Previous attempts to incorporate happiness or wellbeing into biophysical frameworks have consistently introduced a fundamental dimensional inconsistency: psychophysiological variables (e.g., HRV in milliseconds, oxytocin in pg/ml) were treated as additive flux components in kg/day.

This module resolves the issue by redefining wellbeing as a **dimensionless flux quality modulator**, rather than a flux quantity.

Core formulation:

- $EFU_eff = EFU_base \times \eta(W)$
- $\eta(W) = 1 + \alpha \times (W - 0.50)$
- $W = 0.45 \times N_auto + 0.35 \times N_social + 0.20 \times N_cogn$

The index $\eta(W) \in [0.50, 1.20]$ rescales effective biophysical output.

All proxy variables are normalized to [0,1] using empirically grounded functions:

- logarithmic (HRV, social interaction counts)
- inverted-U (cortisol rhythm)
- linear (task completion)

CEWS Integration

Integration into the Civilizational Early Warning System (CEWS) introduces adaptive threshold modulation:

- $Trigger_eff = Trigger_base \times (2 - \eta(W))$

Nonlinear Parasite Coupling

The nonlinear interaction with 600-series antiflux parasites is defined as:

- $P_eff = P_base \times e^{(k \times (1 - \eta(W)))}$ $(k \approx 2.3)$

At $\eta = 0.70 \rightarrow$ parasite effectiveness doubles
At $\eta = 0.50 \rightarrow$ approaches ~5× baseline

This exponential amplification applies across:

- attention extraction systems (600.53)
- ultra-processed food metabolic disruption (600.58)
- synthetic chemical burden tolerance (600.52)

Asymmetry Theorem (050.1.4)

The degradation of $\eta(W)$ occurs **5–10× faster** than its restoration at the behavioral–physiological level, with direct implications for prevention-focused policy design and return on investment (ROI).

MODULE 050.2 — EPIGENETIC FLUX MEMORY INDEX (EFM)

Instantaneous wellbeing efficiency cannot explain persistent low-efficiency states following environmental improvement. Module 050.2 addresses this via the **Epigenetic Flux Memory Index (EFM)** — representing inherited population-level wellbeing capacity.

Three markers were selected based on effect size and measurability; low-impact markers were excluded.

Definition:

- $EFM = 0.40 \times EFM_T + 0.35 \times EFM_C + 0.25 \times EFM_F$

Where:

- EFM_T → telomere length (cumulative stress proxy)
- EFM_C → NR3C1 methylation (HPA-axis sensitivity)
- EFM_F → FKBP5 methylation (trauma inheritance marker)

$EFM \in [0.70, 1.10]$

Integration with Module 050.1

- $\eta(W)_effective = \eta(W)_instantaneous \times EFM$

Epigenetic Floor Theorem (050.2.1)

Each population possesses a minimum epigenetic wellbeing threshold below which sustainable improvement is not possible — only temporary compensation at high energetic cost.

Effectiveness of 700-series regenerative protocols declines exponentially below:

- $EFM = 0.80$

Multi-scale Temporal Asymmetry

Scale	Degradation	Restoration
Behavioral	weeks	months (5–10×)
Physiological	months	years (10–20×)
Epigenetic	decades	generations (50–100×)

Calibration Status

Two empirical constants remain to be calibrated:

- α (η sensitivity): expected 0.05–0.15
- k (parasite exponent): expected 1.80–2.80

Calibration protocols are defined in Modules 050.1.5 and 050.1.6 (Northern European small-state context, ~200 participants, 90-day design).

FILE 3 – Notation Reference Sheet (Polished)

(Itt már nagyon jó volt az eredeti — csak terminológiailag egységesítettem)

Key improvement:

- "Measured / Observed / Corrected" konzisztensen
- "True η(W)" helyett: **Corrected η(W)** (tudományosabb)

FILE 4 – Key Equations Sheet

✔ Már publikációszintű — nem kellett módosítani, csak apró spacing / jelölési egységesítés.

FILE 5 – Upload Checklist (Professionalized)

PRE-UPLOAD VALIDATION

- ORCID identifier available
- All files exported in PDF format

- EZ-compliance verified:
  - No restricted country references
  - Entity_EE naming consistent
  - Internal log (900.5) restricted to headers only
- Related DOIs verified and accessible

UPLOAD SEQUENCE

EFU-W Teljes Modul-csomag

050.1 | 050.1.5 | 050.1.6 | 050.2 | v1.0 VÉGLEGES

EFU SYSTEM v5.1 – ZENODO UPLOAD PACKAGE
Wellbeing Flux Efficiency Index + Epigenetic Memory Index
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Author: István Simor
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Related DOI: 10.5281/zenodo.18888082
10.5281/zenodo.18766850

FÁJL 1 – Zenodo Record Metadata (kitöltési adatlap)

MEZŐ	TARTALOM
Resource type:	Publication → Technical note
Title:	EFU-W: Wellbeing Flux Efficiency Index and Epigenetic Memory Index — Dimensionless Modulators for Biophysical Civilizational Accounting (EFU System v5.1, Modules 050.1–050.2)
Authors:	Simor, István [ORCID: to be inserted]
Publication date:	2026-03-22
Description:	[→ FÁJL 2: Abstract szövege]
Keywords:	embodied flux unit, EFU system, wellbeing quantification, biophysical economics, η(W), epigenetic memory,

telomere length, NR3C1 methylation, FKBP5, HRV normalization, CEWS, civilizational stability, asymmetry theorem, antiparasite resistance, regenerative protocols

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Version: 1.0

Language: English

Related identifiers:
→ Is part of:
github.com/x8cqjnbvnm-hub/EFU-System-v5.1-Diamond
→ Is related to:
DOI 10.5281/zenodo.18888082
[EFU System v5.1 main repository]
→ Is related to:
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[MROI preprint — companion metric]

Funding: Independent research — no institutional funding

Notes: Module 050.3 (integrated η(W)_effective combining 050.1 × 050.2) is in preparation and will be submitted as v1.1 following pilot calibration.

FÁJL 2 – Absztrakt (Zenodo Description mező)

The EFU (Embodied Flux Unit) System quantifies civilizational metabolism through biophysical flux accounting. This technical note presents two interdependent modules that complete the human wellbeing dimension of the framework.

MODULE 050.1 — WELLBEING FLUX EFFICIENCY INDEX η(W)

Prior attempts at integrating happiness or wellbeing into biophysical frameworks have committed a systematic dimensional error: expressing psychophysiological variables (HRV in milliseconds, oxytocin in pg/ml) as additive flux components in kg/day. This module resolves the error by repositioning wellbeing as a flux quality modulator rather than a flux quantity.

Core formula:

$$EFU_eff = EFU_base \times \eta(W)$$
$$\eta(W) = 1 + \alpha \times (W - 0.50)$$
$$W = 0.45 \times N_auto + 0.35 \times N_social + 0.20 \times N_cogn$$

The dimensionless index $\eta(W) \in [0.50, 1.20]$ scales effective biophysical output. All proxy variables are normalized to [0, 1] through empirically grounded functions specified in full (logarithmic for HRV and social interaction counts;

inverted-U for cortisol rhythm; linear for task completion).

CEWS (Civilizational Early Warning System) integration introduces threshold modulation:

$$Trigger_eff = Trigger_base \times (2 - \eta(W))$$

The nonlinear coupling with 600-series antiflux parasites is formalized as:

$$P_eff = P_base \times e^{(k \times (1 - \eta(W)))} \quad k \approx 2.3$$

At $\eta = 0.70$, parasite effectiveness doubles; at $\eta = 0.50$ it approaches fivefold baseline. This exponential structure applies across attention economy extraction (600.53), ultra-processed food metabolic disruption (600.58), and synthetic chemical burden tolerance elevation (600.52).

The Asymmetry Theorem (050.1.4) formalizes the temporal asymmetry: $\eta(W)$ degradation proceeds 5–10× faster than restoration at the behavioral-physiological level, with direct policy implications for prevention ROI.

MODULE 050.2 — EPIGENETIC FLUX MEMORY INDEX (EFM)

Instantaneous $\eta(W)$ cannot explain why population wellbeing efficiency returns to low values even when external conditions improve. Module 050.2 addresses this through the Epigenetic Flux Memory Index — the inherited wellbeing capacity of a population.

Three markers were selected based on effect size and measurability. Markers with negligible independent effect (DRD4 methylation, generic epigenetic clock variants) were explicitly excluded.

$$\begin{aligned} EFM = {} & 0.40 \times EFM_T \quad \text{(telomere length — cumulative population stress proxy)} \\ & + 0.35 \times EFM_C \quad \text{(NR3C1 methylation — cortisol receptor sensitivity, HPA axis heritability)} \\ & + 0.25 \times EFM_F \quad \text{(FKBP5 methylation — trauma inheritance marker)} \end{aligned}$$

$$EFM \in [0.70, 1.10]$$

Integration with 050.1:

$$\eta(W)_effective = \eta(W)_instantaneous \times EFM$$

The Epigenetic Floor Theorem (050.2.1) states that every population has a minimum $\eta(W)_epi$ below which instantaneous wellbeing cannot be sustainably elevated — only temporarily compensated at high energetic cost. The effectiveness of 700-series regenerative protocols decreases exponentially below EFM = 0.80.

The asymmetry extends across three temporal scales:

Behavioral:	degradation weeks, restoration months	(5–10×)
Physiological:	degradation months, restoration years	(10–20×)
Epigenetic:	degradation decades, restoration generations	(50–100×)

CALIBRATION STATUS

Two empirical constants remain pending pilot calibration:
α (η sensitivity): expected 0.05–0.15
k (parasite exponential): expected 1.80–2.80

Calibration protocol and recruitment/ethical framework are specified in companion documents 050.1.5 and 050.1.6 (Entity_EE reference zone, Northern European small-state context, 200-participant design, 90-day protocol).

FÁJL 3 – Jelölési referencialap (teljes)

EFU-W NOTATION REFERENCE — v1.0 FINAL
Modules 050.1 and 050.2

CORE QUANTITIES

Symbol	Definition	Unit	Range
EFU_base	Base biophysical flux	kg/day	≥ 20
EFU_eff	Effective biophysical flux	kg/day	≥ 10
η(W)	Wellbeing efficiency index	—	[0.50, 1.20]
η_mért	Measured η(W)	—	[0.50, 1.20]
η_obs	Observer bias correction	—	[0.02, 0.08]
η_valós	Corrected η(W)	—	[0.50, 1.20]
η_ref	Population median reference	—	1.00
η(W)_effective	Instantaneous × epigenetic	—	[0.50, 1.20]
W	Weighted wellbeing index	—	[0.00, 1.00]
ΔW	Deviation from median	—	[−0.50, +0.50]

THREE-AXIS COMPONENTS (Module 050.1)

N_auto	Autonomic axis index	—	[0.00, 1.00]
N_social	Social axis index	—	[0.00, 1.00]
N_cogn	Cognitive axis index	—	[0.00, 1.00]

NORMALIZED PROXY VARIABLES

HRV_norm	Normalized HRV	—	[0.00, 1.00]
k_norm	Normalized cortisol rhythm	—	[0.00, 1.00]
i_norm	Normalized interactions	—	[0.00, 1.00]
b_norm	Normalized trust index	—	[0.00, 1.00]
c_norm	Normalized physical contact	—	[0.00, 1.00]
f_norm	Normalized flow time	—	[0.00, 1.00]
t_norm	Normalized task completion	—	[0.00, 1.00]

EPIGENETIC QUANTITIES (Module 050.2)

EFM	Epigenetic Flux Memory Index	—	[0.70, 1.10]
EFM_T	Telomere length component	—	[0.00, 1.00]
EFM_C	NR3C1 methylation component	—	[0.00, 1.00]
EFM_F	FKBP5 methylation component	—	[0.00, 1.00]

CALIBRATION CONSTANTS

α	η(W) sensitivity constant	—	[0.05, 0.15]
k	Parasite exponential const.	—	[1.80, 2.80]

CEWS QUANTITIES

Trigger_base	CEWS base threshold	sys. unit fixed	
Trigger_eff	Modulated CEWS threshold	sys. unit variable	
P_base	Baseline parasite effect	—	reference
P_eff	Effective parasite effect	—	[0.6×, 4.5×]

PILOT QUANTITIES

η_obs	Observer bias estimate	—	D-group
SR	Subjective Resonance	WHR 0–10	observed

AXIS WEIGHTS (fixed, v1.0)
N_auto: 0.45
N_social: 0.35
N_cogn: 0.20

PROXY WEIGHTS WITHIN AXES (fixed, v1.0)
N_auto: HRV_norm 0.60 | k_norm 0.40
N_social: i_norm 0.50 | b_norm 0.30 | c_norm 0.20
N_cogn: f_norm 0.60 | t_norm 0.40

EFM WEIGHTS (fixed, v1.0)
EFM_T: 0.40 | EFM_C: 0.35 | EFM_F: 0.25

FÁJL 4 – Kulcsegyenletek lap (ábraszöveg-specifikáció)

EFU-W KEY EQUATIONS — Textual Figure Specification
For typesetting / LaTeX rendering

EQUATION 1 — Core modulator
$EFU_{eff} = EFU_{base} \times \eta(W)$

EQUATION 2 — η(W) definition
$\eta(W) = 1 + \alpha \times (W - 0.50)$

EQUATION 3 — W composition
$W = 0.45 \times N_{auto} + 0.35 \times N_{social} + 0.20 \times N_{cogn}$

EQUATION 4 — N_auto
$N_{auto} = 0.60 \times HRV_{norm} + 0.40 \times k_{norm}$

EQUATION 5 — N_social
$N_{social} = 0.50 \times i_{norm} + 0.30 \times b_{norm} + 0.20 \times c_{norm}$

EQUATION 6 — N_cogn
$N_{cogn} = 0.60 \times f_{norm} + 0.40 \times t_{norm}$

EQUATION 7 — HRV normalization
$HRV_{norm} = \log(HRV_{ms} / 20) / \log(4)$
[floor: 0.00 at HRV < 20 ms; cap: 1.00 at HRV > 80 ms]

EQUATION 8 — Cortisol rhythm normalization
$k_norm = 1 - |\ln(R/E) - \ln(3)| / \ln(4)$
[R/E = morning/evening cortisol ratio; optimum at R/E = 3.0]

EQUATION 9 — Interaction normalization
$i_norm = \ln(encounters + 1) / \ln(7)$
[cap: 1.00 at ≥ 6 encounters/day]

EQUATION 10 — Trust normalization
$b_norm = (score - 20) / 50$
[Rosenberg Trust Scale; floor: 0.00 at score < 20]

EQUATION 11 — Contact normalization
$c_norm = \min(events / 4, 1.00)$

EQUATION 12 — Flow time normalization
$f_norm = \min(\ln(minutes + 1) / \ln(241), 1.00)$

EQUATION 13 — Task completion normalization
$t_norm = value / 3$ [0–3 self-report scale]

EQUATION 14 — Observer bias correction
$\eta_valós = \eta_mért - \eta_obs$

EQUATION 15 — CEWS threshold modulation
$Trigger_eff = Trigger_base \times (2 - \eta(W))$

EQUATION 16 — Parasite nonlinear coupling
$P_eff = P_base \times e^{(k \times (1 - \eta(W)))}$

EQUATION 17 — EFM composition (Module 050.2)
$EFM = 0.40 \times EFM_T + 0.35 \times EFM_C + 0.25 \times EFM_F$

EQUATION 18 — Integrated effective η
$\eta(W)_effective = \eta(W)_instantaneous \times EFM$

EQUATION 19 — CEWS with epigenetic layer
$Trigger_eff = Trigger_base \times (2 - \eta(W)_effective)$

FÁJL 5 – Feltöltési ellenőrzőlista és Feltöltési Sorrend

PRE-UPLOAD ELLENŐRZÉS

☐ ORCID azonosító kész
☐ Minden fájl PDF formátumban
☐ EZ-compliance ellenőrzés:
→ Nincs tiltott ország neve egyetlen fájlban sem
→ Entity_EE referencia konzisztens
→ Belső log: 900.5 csak fejlécben, nem főszövegben
☐ Kapcsolódó DOI-k ellenőrizve (zenodo.18888082, zenodo.18766850 elérhetők)

FELTÖLTÉSI SORREND

LÉPÉS	TEVÉKENYSÉG	IDŐ

1. zenodo.org → New upload 1 perc
2. Resource type: Publication →
   Technical note 1 perc
3. Fájlok feltöltése:
   EFU-W_050-1_core_v1.0.pdf
   EFU-W_050-1-5_calibration_v1.0.pdf
   EFU-W_050-1-6_recruitment_v1.0.pdf
   EFU-W_050-2_epigenetic_v1.0.pdf
   EFU-W_summary_v1.0.pdf
   EFU-W_notation_v1.0.pdf
   EFU-W_equations_v1.0.pdf 5 perc
4. Metadata kitöltése (FÁJL 1 alapján) 5 perc
5. Description beillesztése
   (FÁJL 2 szövege) 2 perc
6. SAVE DRAFT → DOI generálódik azonnal

*** DOI MEGÉRKEZIK ***
Pl.: 10.5281/zenodo.XXXXXXXX

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7. DOI visszaírása minden fájlba
   (fejléc + citation mező) 10 perc
8. Fájlok újragenerálása PDF-be 5 perc
9. Régi fájlok törlése Zenodo draftból
10. Frissített fájlok feltöltése 3 perc
11. PUBLISH → végleges időbélyeg 1 perc

Teljes idő: ~35 perc

POST-UPLOAD (D+0):
- ☐ GitHub README frissítése DOI badge-dzsel
- ☐ EFU-W modulok hozzáadva a repo struktúrához
- ☐ Belső log: 050.1–050.2 státusz → PUBLISHED

POST-UPLOAD (D+3):
- ☐ Mini-pilot előkészítés indul
- ☐ NR3C1 nyálminta protokoll döntés (FC)

Összefoglaló – A Csomag Tartalma

FÁJL TARTALOM	STÁTUSZ
1. Zenodo metadata adatlap	☐ KÉSZ
2. Abstract / Description szöveg	☐ KÉSZ
3. Notation reference sheet	☐ KÉSZ
4. Key equations sheet	☐ KÉSZ
5. Upload checklist + sorrend	☐ KÉSZ

+ Modulok (előző munkamenetek):

050.1 Core η(W)	☐ FINAL
050.1.5 Calibration protocol	☐ FINAL
050.1.6 Recruitment & ethics	☐ FINAL
050.2 Epigenetic Memory Index	☐ FINAL

ZENODO SUBMISSION PACKAGE: COMPLETE
EFU System v5.1 — Modules 050.1 through 050.2

Fire Chief Override: István Simor
EZ-compliance: ACTIVE
2026-03-22